CO2xNxPhrag: Porewater Methane

Month Year Samples

2025-08-22

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##Run Information

```
cat("Run Information: Brenden D. Blakley ") #lets you know what section you're in
```

```
## Run Information: Brenden D. Blakley
```

```
#set the run date & user name
run_date <- "09/02/2025"
sample_year <- "2025"
sample_month <- "July"
user <- "Brenden Blakley"
#identify the files you want to read in
#read in as a list to accommodate multiple runs in a month

GHG_file <- "Raw Data/20250905_GHGdata_LTREB_PHRAG.csv"


# Define the file path for QAQC log file - NO Need to change just check year
log_path <- "Processed Data/LTREB_Phrag_ShimadzuGC_QAQC_2025.csv"

# Define final path for data be sure to change year and month and run number
final_path <- "Processed Data/GCReW_CO2xNxPhrag_Porewater_CH4_202507_2.csv"

#Gas in Water Calculations
# "TempC" column value in Celsius (equilibration temperature) and volume of gas (in liters) in syringe
TempC = 22.7 #Equilibration Temperature (Celsius)

volume_g = 0.015 #volume of gas (in liters) in syringe when equilibrized
volume_w = 0.015 #volume of water (in liters) in syringe when equilibrized (should be equal to volume_g

#record any notes about the run or anything other info here:
run_notes <- ""



#Collection_Dates = "Raw Data/Sample_Collection_Dates_2024.csv"

cat(run_notes)
```

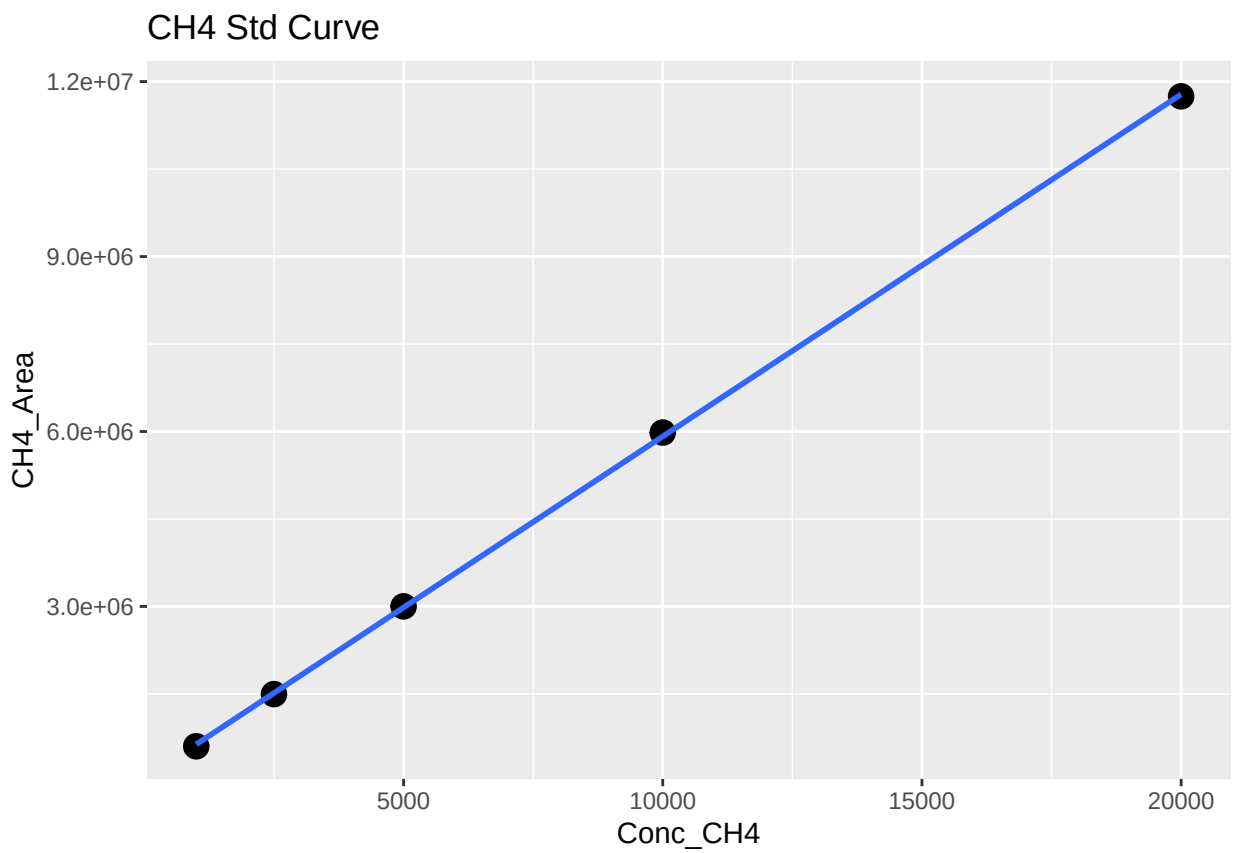
##Set Up

##Read in metadata and create similar sample IDs for matching to samples

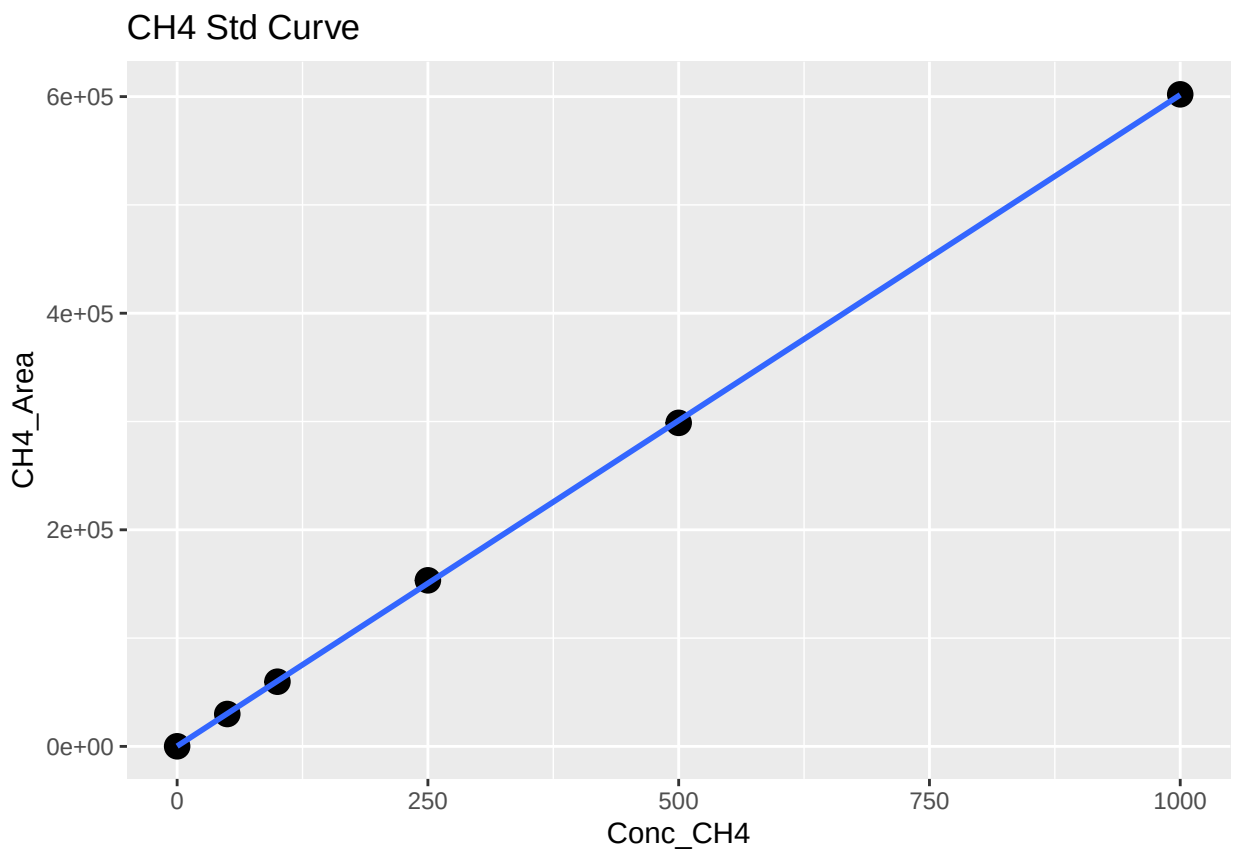
1 Read in data files

2 Assess standard curves

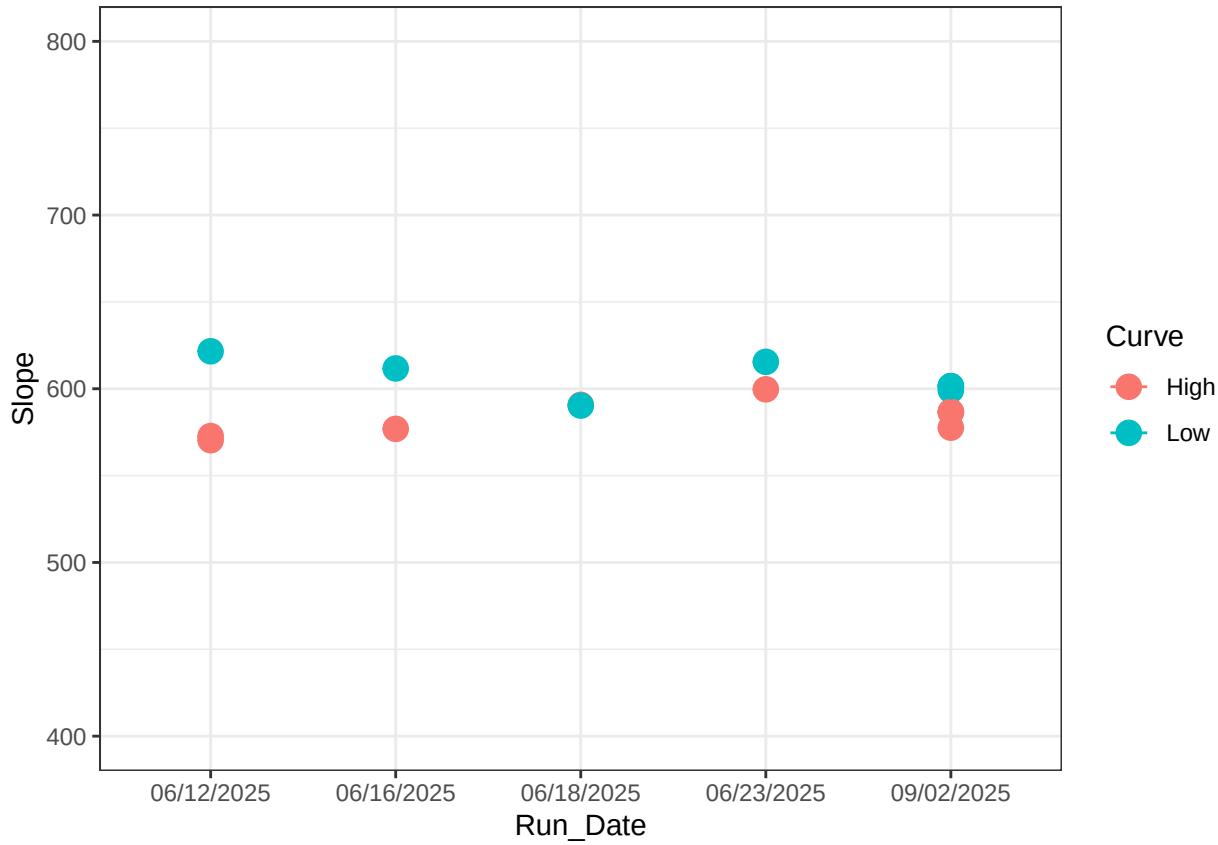
```
## `geom_smooth()` using formula = 'y ~ x'
```




```
## `geom_smooth()` using formula = 'y ~ x'
```




```
##   X Curve        R2    Slope    Intercept   Run_Date
## 1 1  High 0.9986295 576.8990  173986.2174 06/16/2025
## 2 2   Low 0.9990260 611.6961   -3791.2684 06/16/2025
## 3 3  High 0.9999101 570.3925  138747.3478 06/12/2025
## 4 4   Low 0.9979098 621.5734    3224.2535 06/12/2025
## 5 5  High 0.9989882 590.7374 -109278.7391 06/18/2025
## 6 6   Low 0.9990995 590.3138     954.7682 06/18/2025
```



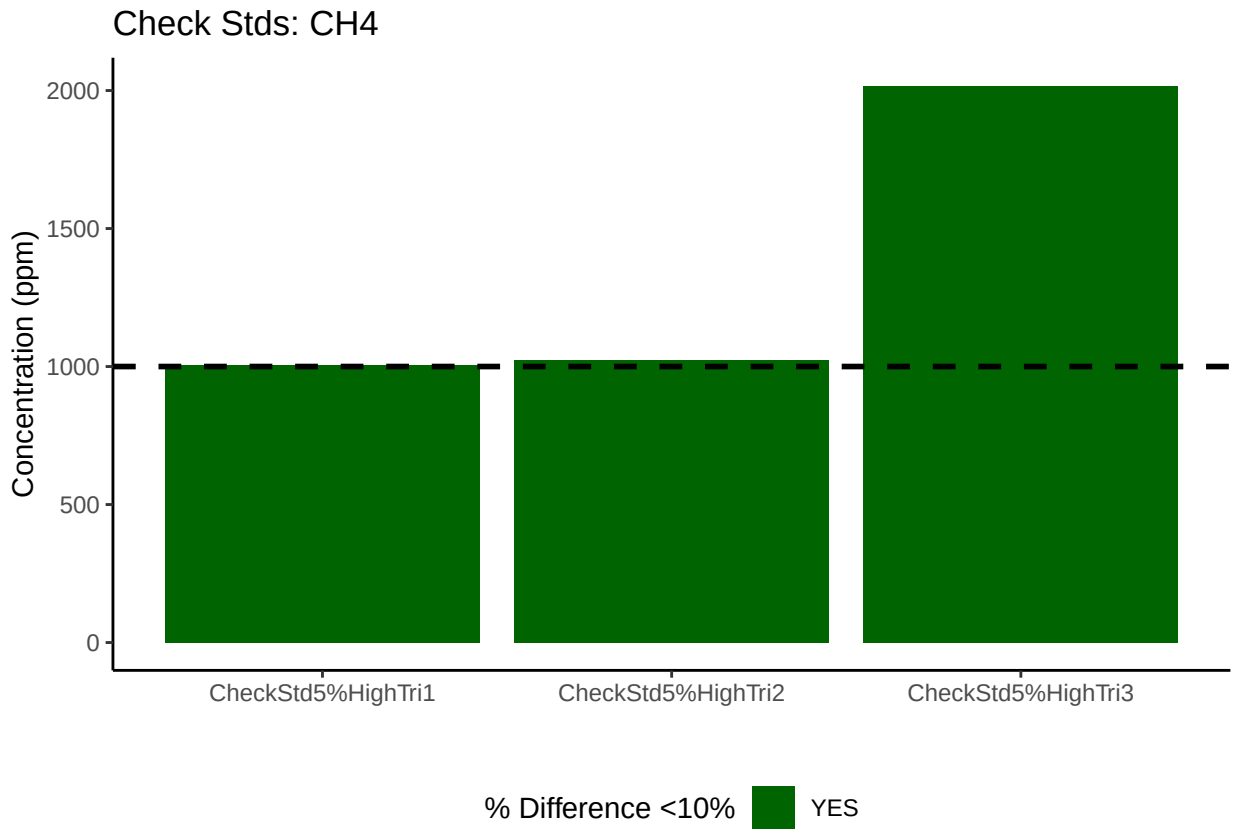

```
## High_R2 Good
## Low_R2 Good
```

2.1 Now calculate the CH4 & CO2 concentrations in ppm

3 Anlayze the Check Standards

```
## >60% of CH4 Check Standards are within range of expected concentration - PROCEED
```

```
## Warning: Using ‘size‘ aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use ‘linewidth‘ instead.
## This warning is displayed once every 8 hours.
## Call ‘lifecycle::last_lifecycle_warnings()‘ to see where this warning was
## generated.
```



4 Analyze the Blanks

```
## Assess Blanks
```

```
## >60% of CH4 Blanks are below the lower 25% quartile of samples - PROCEED
```

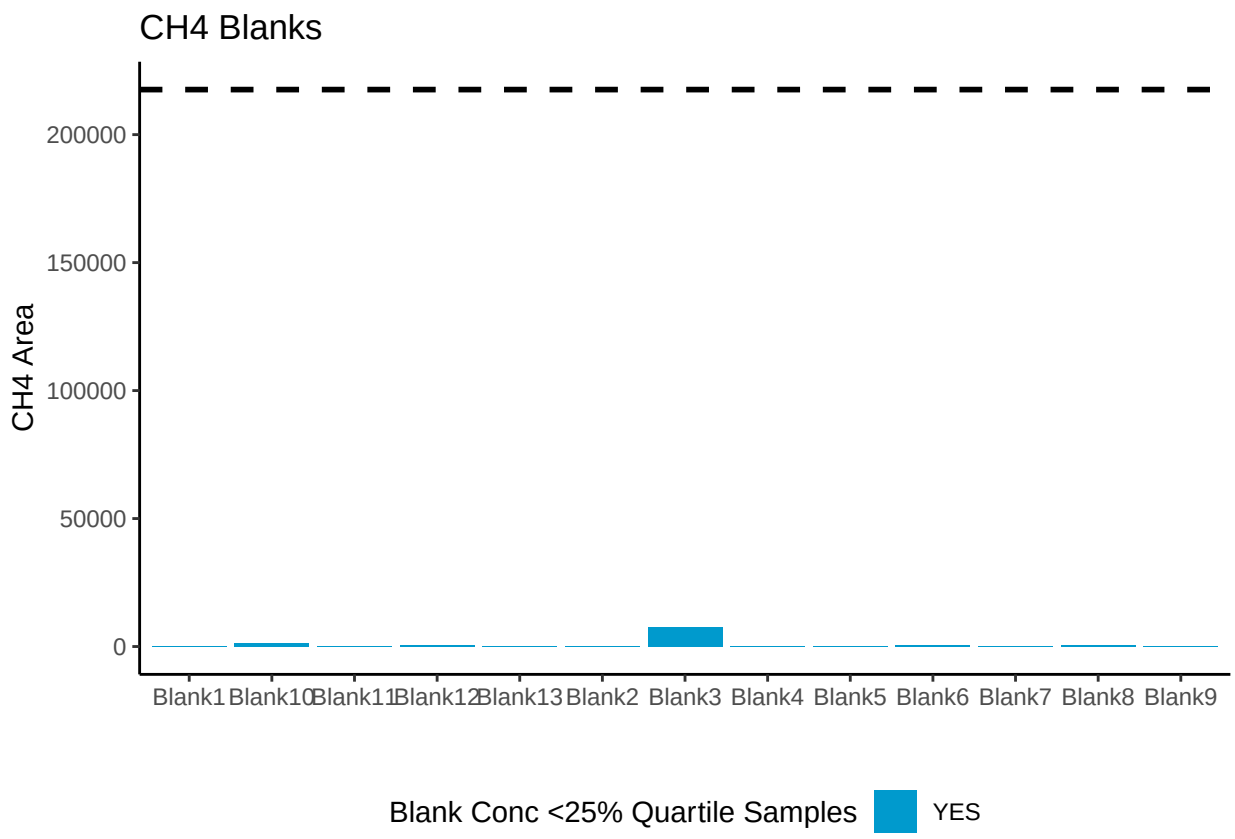
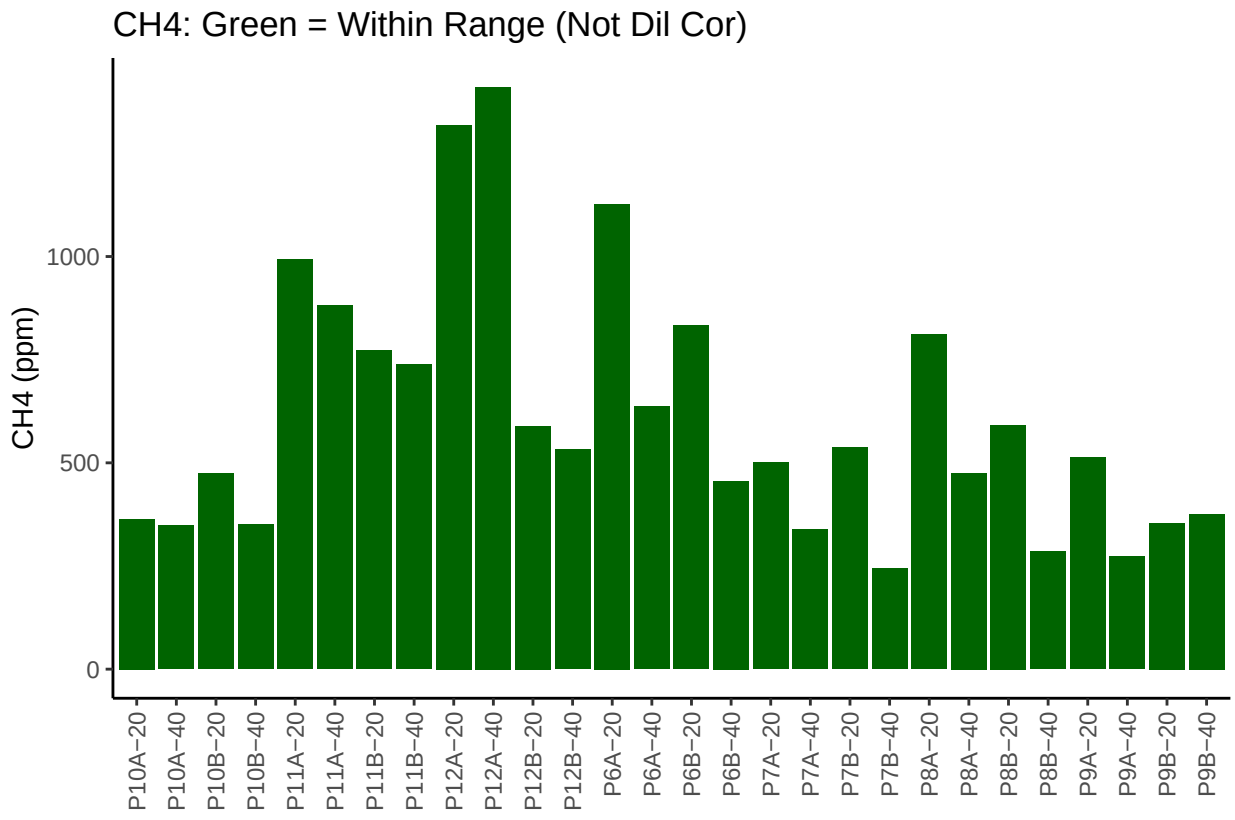
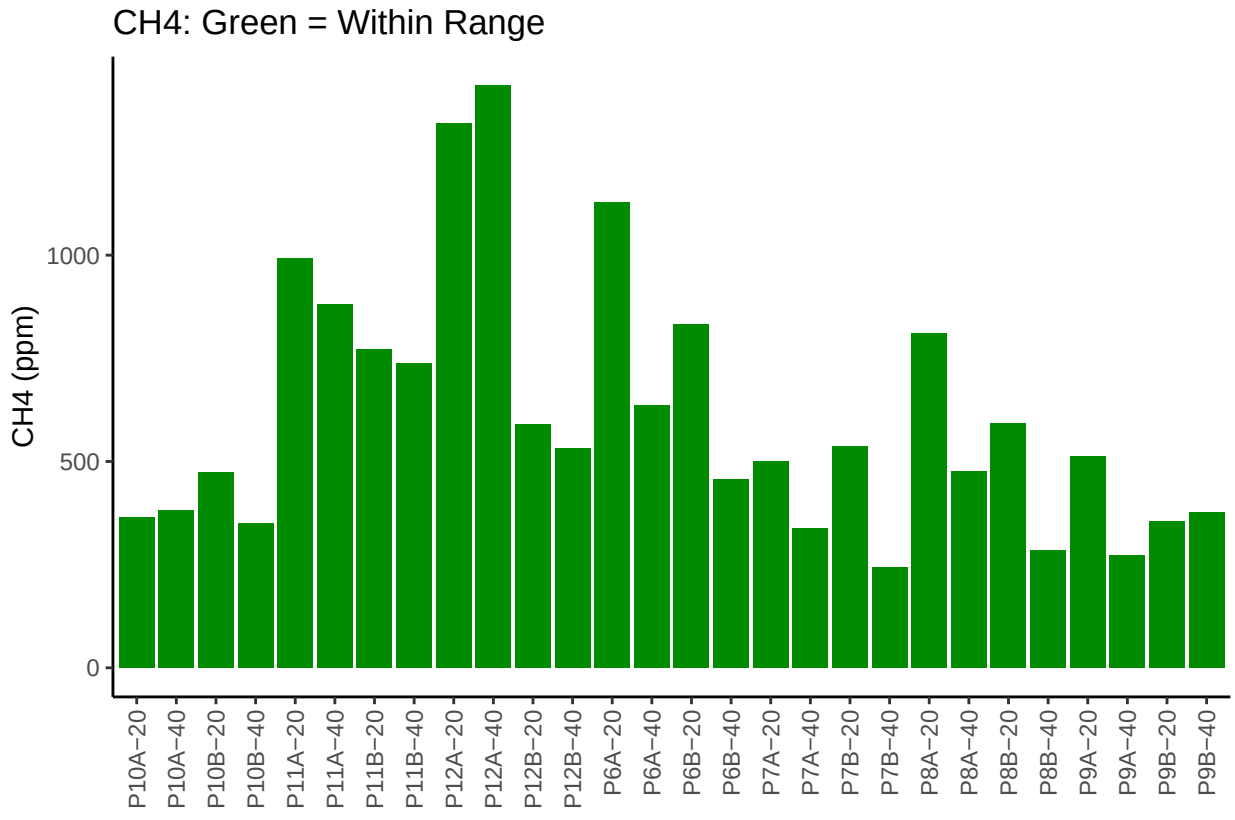


Table 1: Mean Area of Blanks

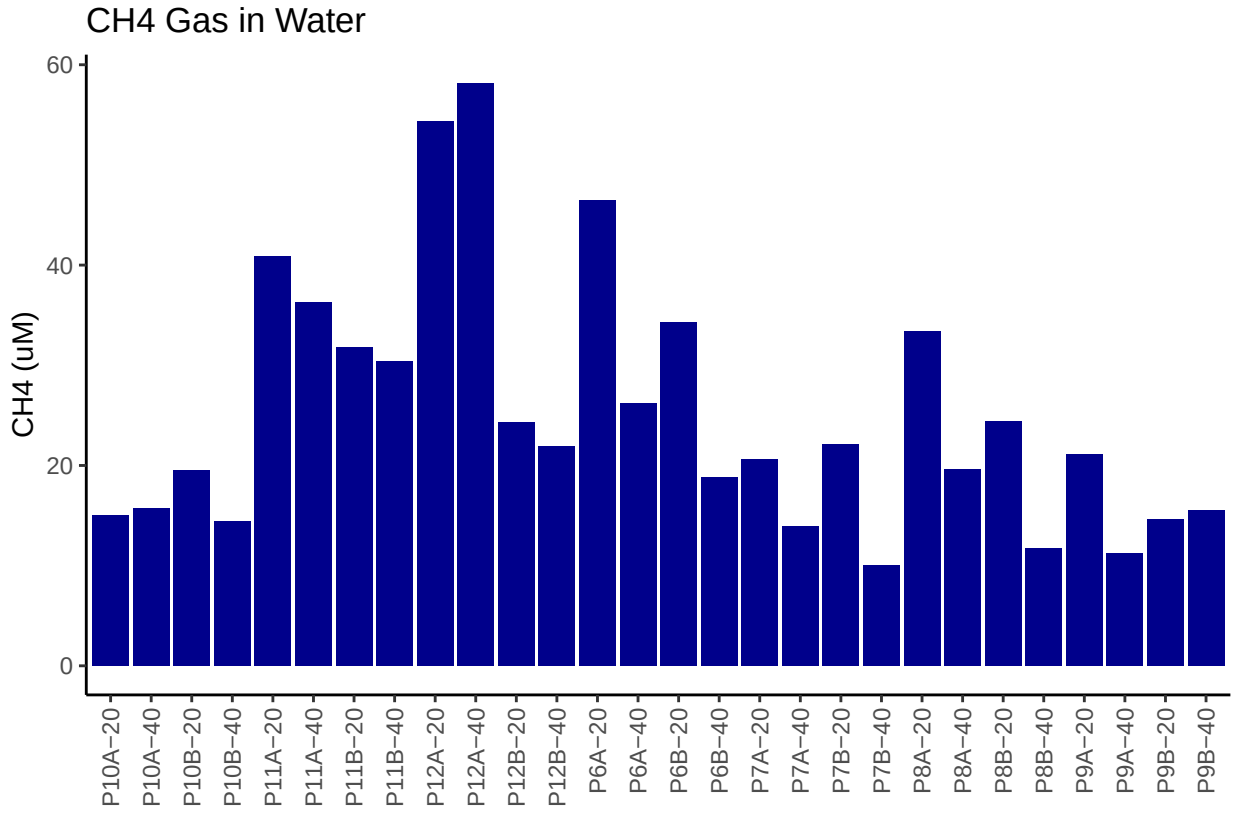
Test	Blank_Mean_Area
CH4	977.6923

5 Dilution correct samples and Check Range





6 Calculate gas in water



7 Merge samples with Metadata & check all samples are present

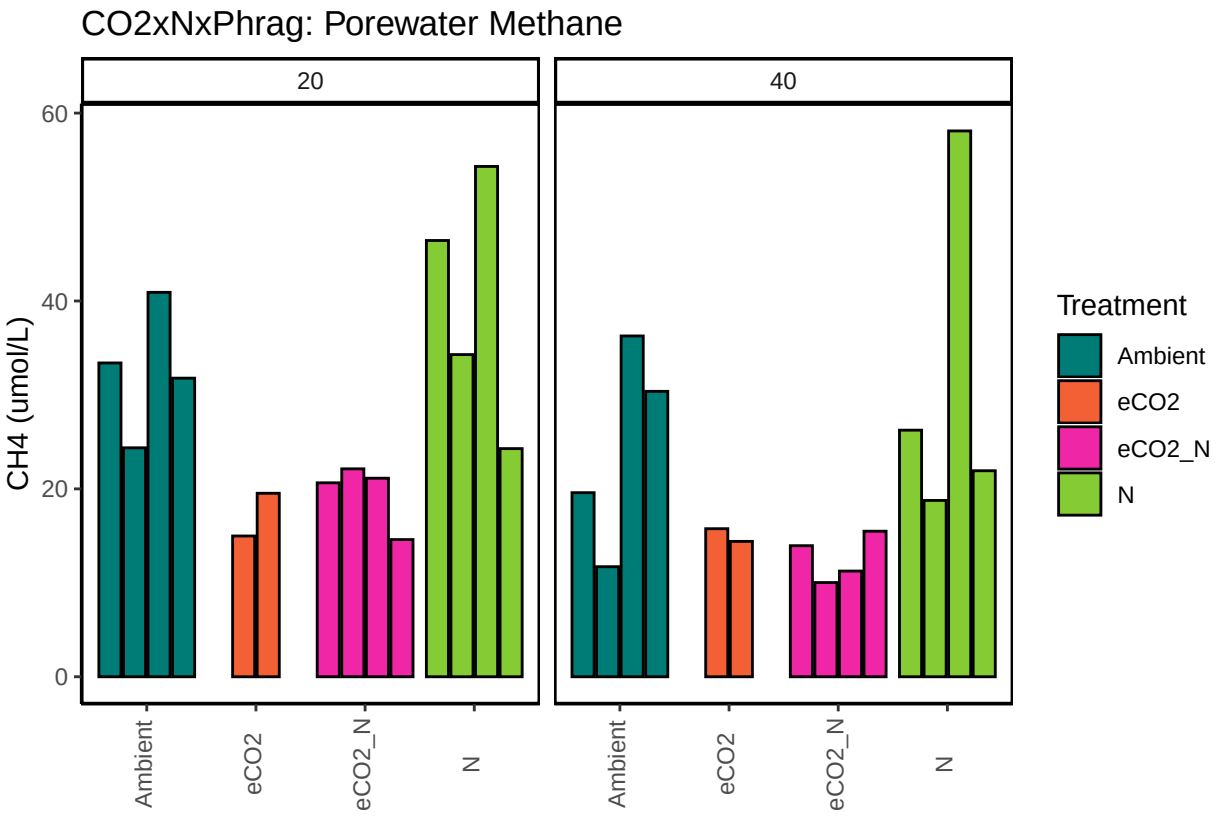
```
## Merge Samples with Metadata
```

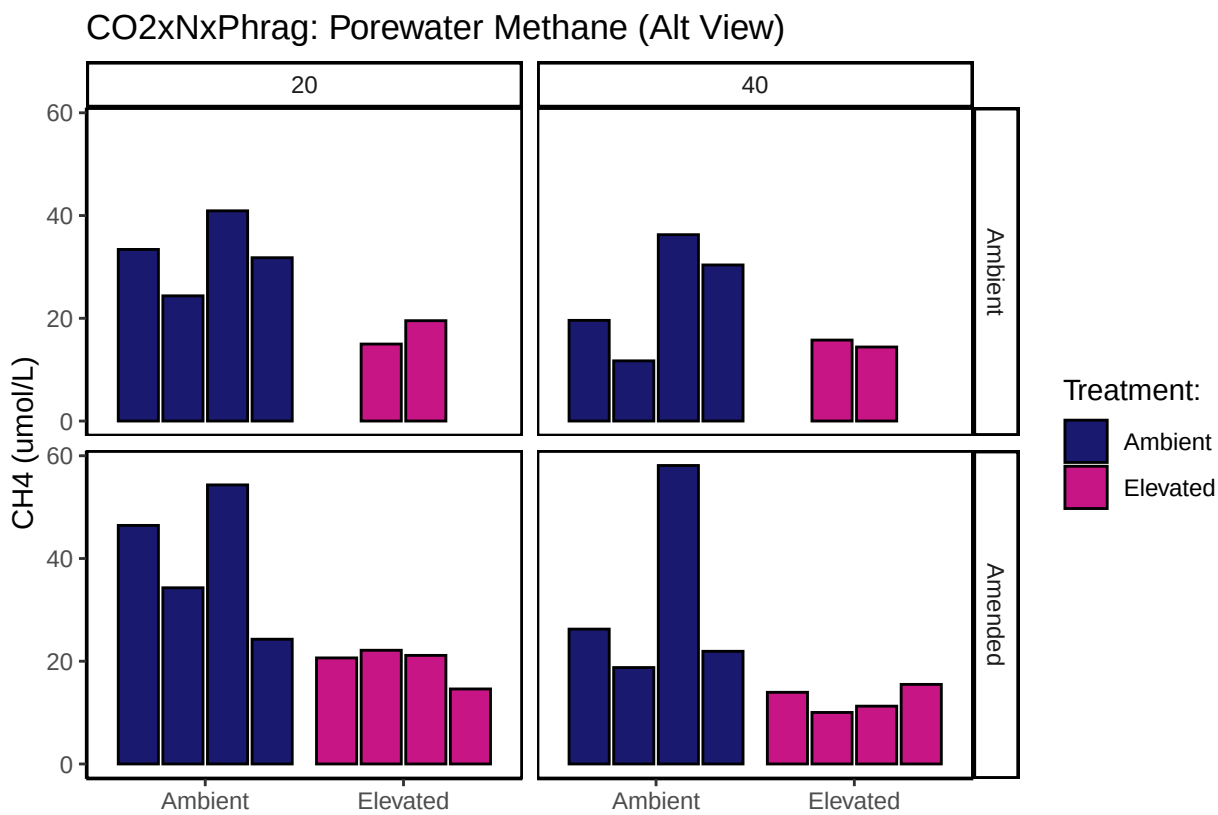
```
## Some sample IDs are missing from data.
```

```
##  [1] "P1A-20" "P1A-40" "P1B-20" "P1B-40" "P2A-20" "P2A-40" "P2B-20" "P2B-40"
##  [9] "P3A-20" "P3A-40" "P3B-20" "P3B-40" "P4A-20" "P4A-40" "P4B-20" "P4B-40"
## [17] "P5A-20" "P5A-40" "P5B-20" "P5B-40"
```

8 Visualize Data

```
## Visualize Data
```


CO2xNxPhrag: Porewater Methane
20
40
60
40
20
0
CH4 (umol/L)
Ambient
eCO2
eCO2_N
N
Ambient
eCO2
eCO2_N
N
Treatment
Ambient
eCO2
eCO2_N
N

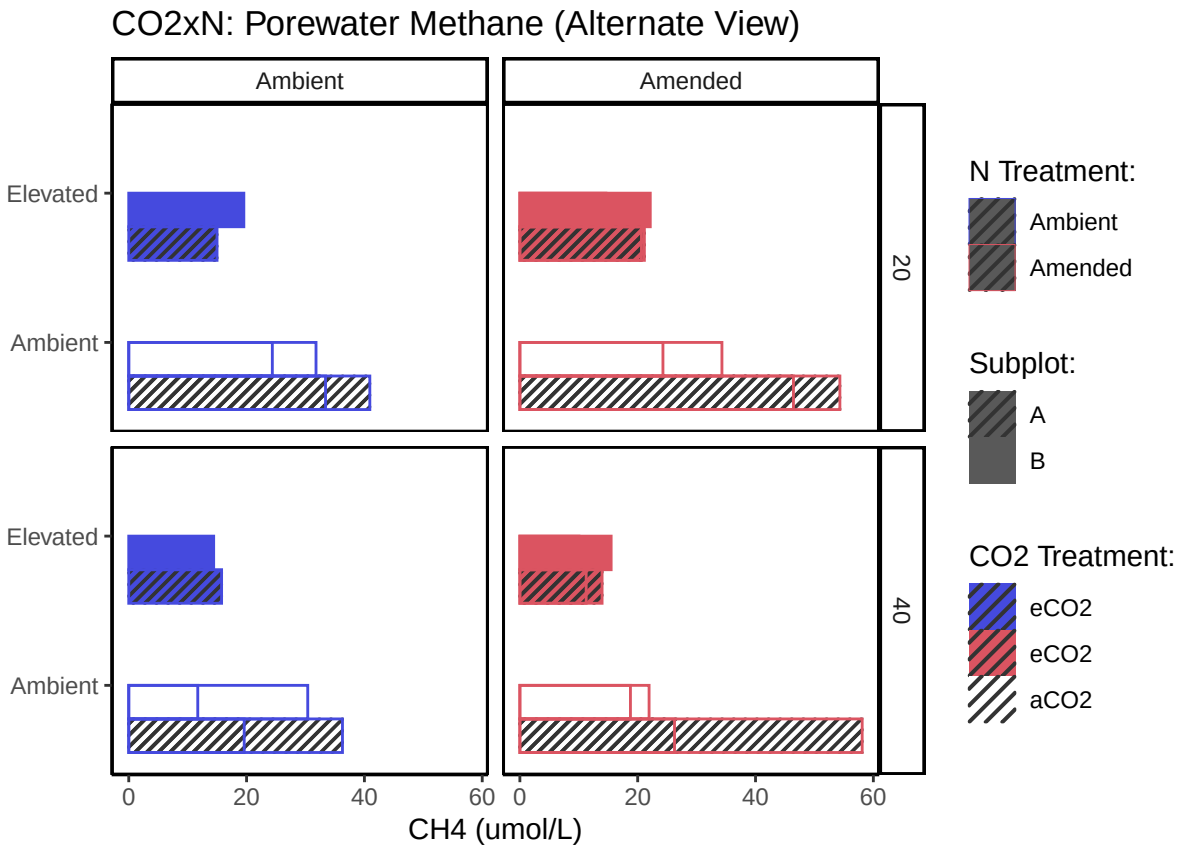



```
CH4_forplot <- CH4_forplot %>%
  mutate(Treatment = ifelse(
    CO2_Treatment == "Elevated" &
      N_Treatment == "Amended",
    "eCO2_N",
    ifelse(
      CO2_Treatment == "Elevated" &
        N_Treatment == "Ambient",
      "eCO2",
      ifelse(CO2_Treatment == "Ambient" &
               N_Treatment == "Amended"  , "N" , "Ambient")
    )
  ))



viz_CH4_plot2 <- ggplot(data = CH4_forplot, aes(
  x = as.factor(CO2_Treatment),
  y = CH4_Conc_uM,
  fill = as.factor(Treatment),
  color = as.factor(N_Treatment),
  pattern = as.factor(SubPlot)
)) +
  geom_bar_pattern(
    stat = "identity",
    position = position_dodge(preserve = "single"),
```

```r
                    pattern_fill = "black",
                    pattern_angle = 45,
                    pattern_density = 0.1,
                    pattern_spacing = 0.025,
                    pattern_key_scale_factor = 0.6
  ) +
  facet_grid(Depth_cm ~ N_Treatment, scales = "fixed") +
  scale_fill_manual(
    values = c(
    "eCO2_N" = "#DB5461",
    "eCO2" = "#454ADE",
    "N" = "white",
    "Ambient" = "white"
  ),
    breaks = c("eCO2", "eCO2_N", "Ambient"),
    labels = c("eCO2","eCO2","aCO2")
  ) +
  scale_color_manual(values = c(
    "Ambient" = "#454ADE",
    "Amended" = "#DB5461"
  )) +
  scale_pattern_manual(values = c(A = "stripe", B = "none")) +
  theme_classic() +
  labs(x = " ",
       y = "CH4 (umol/L)",
       title = "CO2xN: Porewater Methane (Alternate View)",
       fill = "CO2 Treatment:",
       color = "N Treatment:",
       pattern = "Subplot:") +
  theme(legend.position = "right") +
  scale_x_discrete(drop = TRUE) +
  theme(
    axis.text.x = element_text(angle = 0, hjust = 0.5),
    panel.border = element_rect(
      color = "black",
      fill = NA,
      linewidth = 0.8
    ),
      legend.key = element_rect("black")
  ) +
coord_flip()
print(viz_CH4_plot2)
```



8.1 Export Processed Data

```
## [1] "09/02/2025"
```

#end